

AN ECLECTIC TECHNOLOGIES FOUNDATIONAL DEFINITION

FD-BL — The Bright Line

Foundational Definition: The Placement Criterion, Its Granularity, Its Scope of Application, and the Obligation It Creates

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Designation: FD-BL **Family:** Foundational Definitions **Version:** 0.3 — DRAFT (§6 open items collapsed to resolved pointers: §6.1→FD-BL-D1, §6.2→FD-BL-D2, §6.3→FD-EV; supersedes v0.2) **Date:** August 2026 **Author:** Tim Roxey **Organization:** Eclectic Technologies **Audience:** Standards committee reviewers; regulatory agency technical staff; engineering staff at regulated utilities and industrial facilities; suppliers of ICS/OT products incorporating machine learning or generative AI **Parent:** None. This instrument defines. See §2. **Companion root definition:** FD-LD (layer decomposition and determinism)

1. Purpose and the Defect This Instrument Corrects

The Bright Line is the central concept of this corpus. Roughly forty published instruments inherit it. Sector frameworks, facility class documents, vendor conformance obligations, and regulatory annexes all resolve to it.

It is currently stated five different ways, and the five are not compatible.

They are not five competing answers to one question. They are answers to four different questions, each mistaken for the whole. A reader arriving at any one instrument acquires a definition that will not survive contact with the next, and a determination made against one formulation does not bind the others.

The underlying cause is worth naming, because it governs how this instrument is written. The corpus never stated what *kind of thing* the Bright Line is. Each formulation rests on a different unstated answer to that question — a property of components, a boundary between architectural layers, a threshold on actions — and each is internally coherent given its own unstated premise. The premises were never written down, so they were never examined.

This instrument separates the four questions, answers each once, states its own ontological commitment before defining anything, and establishes which document owns the answer.

2. Authority, Family, and Convention

2.1 Precedence

This instrument owns the definition of the Bright Line for all documents in the corpus.

Foundational Definitions take precedence over sector frameworks, facility class documents, application guides, vendor companions, annexes, and implementation guidance. A sector framework defines *within* a concept; it does not define the concept. This is a tier rule and applies to every foundational definition and every sector framework, not only to the instruments named in this document.

Every other statement of the Bright Line is a restatement. A restatement may be expressed in sector vocabulary and may apply the definition to sector-specific actions. It shall not vary the placement criterion, the granularity, the scope of application, or the obligation. Where an existing instrument varies, this instrument governs and the varying text shall be conformed at that instrument's next revision.

Where an instrument propagates the definition to a series or to subordinate documents, it shall propagate by reference to this instrument rather than by restating text. An instrument that rebroadcasts its own wording will re-introduce variance at every revision cycle.

2.2 The Foundational Definitions family

FD-BL is a member of the Foundational Definitions family. A concept belongs in this family when the correctness of other documents depends on agreeing what kind of thing it is. A concept with a settled external meaning, on which no document's correctness turns, does not belong here however important it is.

FD-LD, covering layer decomposition and determinism, is a companion root definition and is the source text for the scope of application at §4.4. FD-LD owns the determinism and layer vocabulary; it does not own placement, which is defined here.

2.3 Declaration convention

Each instrument in this family shall state what kind of thing its concept is before defining it. Placement of that statement is at the head of the definitional section.

This convention exists because its absence is the identified cause of the defect at §1. A document whose subject is components will tend to state a concept as a property of components without noticing it has done so. Requiring the declaration makes the commitment visible and therefore reviewable.

2.4 What this instrument does not alter

Nothing here alters the criticality band model, the consequence-severity categories, the Command Broker qualification requirements, or the assessment methodology. Those are defined elsewhere and are inherited unchanged.

This instrument is definitional. It states requirements about the definition and about conformance to it. It does not state operational requirements for the systems it describes.

3. The Four Questions

Precision requires that these be kept apart. Nearly every defect in the current text is a collapse of one into another.

Placement. On what property is the line drawn — what makes an action fall above it rather than below it?

Granularity. At what unit is placement determined — the system, the application, or the individual action?

Scope. Which components are subject to Bright Line analysis at all?

Obligation. What does an above-line placement require?

Placement and scope are the pair most often conflated, and that conflation produces the most consequential errors in the current corpus. The determinism property answers the scope question. It does not answer the placement question. A framework that uses determinism to decide where the line falls has made a component's technology the measure of a plant's consequence, which is the reasoning this corpus exists to displace.

4. The Definition

4.1 What kind of thing the Bright Line is

The Bright Line is a threshold on actions.

It is not a property of components. It is not a boundary between architectural layers. It is not a classification of technologies. It is a threshold, and what it partitions is the set of actions a system can take.

Components enter the analysis in one respect only: their determinism decides whether the actions they produce are subject to the threshold. That is the scope question at §4.4, and it is a separate question from where the threshold falls.

Everything else in this instrument follows from this paragraph.

4.2 The placement criterion is consequence

The Bright Line is drawn on the consequence of the action. It is not drawn on the technology that produced the action, on the sophistication of that technology, on whether the action is characterized as advisory or as control, or on the architectural layer in which the producing component resides.

An action falls above the Bright Line when its incorrect performance could produce a consequence at or above the severity threshold established for the facility. It falls below when it could not.

Three corollaries follow, and each corrects an error presently in circulation.

A supplier's characterization of a product does not determine placement. A component described as advisory, as decision support, as optimization, or as operator-free autonomy is placed by what its actions can cause, not by what it is called.

Placement does not follow architectural layer. An action originating in a component described as advisory may be above the line if its consequence is above the threshold. An action originating in a control-layer component may be below the line if its consequence is not. The advisory-to-command seam is a useful place to build enforcement; it is not the definition of the boundary. Building a single gate there on the assumption that it coincides with the line will leave above-line actions ungated on the advisory side and gate below-line actions unnecessarily on the command side.

Placement does not follow determinism. Whether a component is deterministic governs whether it is subject to this analysis, which is §4.4. It does not govern where the line falls.

4.3 Placement is determined action by action

The unit of placement is the individual action, not the system and not the application.

A single product, deployed once, in one facility, will typically produce actions on both sides of the line. A generative advisory system that drafts maintenance work orders, recommends setpoint adjustments, and summarizes historian data produces actions whose consequences differ by orders of magnitude.

Classifying the product as a whole is imprecise in both directions: it imposes interposition on trivial actions, which produces workarounds, and it obscures which specific actions actually require it.

The practical consequence for suppliers is that the required architecture is a consequence classifier that routes action by action, not a single gate at a seam. The classifier's routing decisions, and the basis on which each action was classified, are part of the evidentiary record.

Placement is distinct from criticality band assignment, and the two operate at different units. Band assignment is a property of the application, derived from its consequence-severity profile, and establishes the rigor of assessment, qualification, and audit that applies. Placement is a property of each action within that application and establishes whether the action may proceed without human authorization. A high-band application contains below-line actions. A low-band application may contain above-line ones. Neither determines the other, and an instrument that treats band assignment as placement has answered the granularity question wrongly.

4.4 Scope of application: the determinism carve-out

Bright Line analysis and the Command Broker obligation apply to generative and other non-deterministic components. They do not apply to components that are deterministic within a stated input bound and formally testable in the deployed configuration. FD-LD supplies the underlying vocabulary.

The carve-out exists because a substantial body of machine learning already in service in critical infrastructure — anomaly detection, predictive maintenance, condition-based monitoring, pattern recognition — is deterministic within bounded inputs, produces bounded and testable outputs, and is governed by the existing standards ecosystem. Extending non-deterministic assessment to it would impose unworkable requirements on tools that are functioning safely.

The carve-out is a property test and binds in both directions. A component is not exempted by being called machine learning. The test is whether it is deterministic within a stated input bound and formally testable as deployed. A model whose outputs are not reproducible under identical inputs is a non-deterministic component whatever its architecture is named, and any component relying on the carve-out shall have its input bound stated and verified as part of assessment.

The carve-out shall not be narrowed to enumerated roles. An instrument that permits deterministic machine learning to operate autonomously in named roles has converted a property test into a use-case list, which will exclude conforming components for reasons unrelated to their behavior and will invite relabeling by components that do not conform.

The carve-out is not the line. Determinism decides whether a component is subject to Bright Line analysis. Consequence decides where the line falls for the actions that component produces. A formulation that places deterministic machine learning below the line and generative systems above it has fused the two questions and produces two errors at once: it implies that every generative action is above the line, including those with no material consequence, and it implies that a deterministic component can never produce an above-line action, which is false and which removes from analysis the automated systems most capable of causing harm.

Scope of the carve-out. The carve-out removes a component from *this* obligation. It is not a grant of ungoverned autonomy, and it is not a finding that the component's actions are below the line. Verification and qualification obligations for deterministic components arise elsewhere: in the functional safety ecosystem, in the industrial automation security ecosystem, and for nuclear digital instrumentation and control in the safety system design and qualification standards applicable to that sector. (*Specific standard and clause references to be confirmed against current editions before issue — see §8.*)

This instrument records a gap rather than papering it. Deterministic machine learning taking autonomous, consequential control action — automated dispatch, voltage regulation, load allocation — sits in a partially covered area of that existing ecosystem. The coverage is stronger for conventional deterministic protective functions, which have decades of settled treatment, than for machine-learned

components that satisfy the property test. Closing that gap is outside the scope of this instrument and outside the current scope of this corpus, and is identified here so that the carve-out is not read as an assertion that no gap exists.

The carve-out does not extend to actions taken upon carved-out components. A component within scope of this instrument that modifies a protective function's setpoints, trip logic, bypass state, or actuation criteria is taking an action upon that function. Such actions are placed on their own consequence under §4.2, and they are among the highest-consequence actions available in any facility. The carve-out covers a carved-out component's own actions. It does not cover in-scope components acting upon it.

4.5 The obligation above the line

An action placed above the Bright Line requires human authorization that is substantive and attributable. Substantive means the authorizing human has the information, competence, time, and authority to withhold authorization on the merits. Attributable means the authorization resolves to a named qualified individual and is reconstructable from the audit record.

The authorizing human is the Command Broker. Broker competency, onboarding, and accountability are established in the qualification standard and are not restated here.

The obligation is architectural, not procedural. The control system shall be incapable of executing an above-line action originating from a component within scope of §4.4 without the required authorization. The enforcement mechanism shall not be defeasible by configuration change, software update, or system integration, and the requirement is not discharged by a procedure instructing an operator to confirm.

Accountability for the authorized action rests with the named human broker and with the asset owner-operator. A supplier may build and operate the enforcement mechanism; a supplier may not hold the accountability, and no software component holds it.

Why the obligation attaches where it does. The Command Broker is compensation for unverifiability. It is not a substitute for automation, and it is not a general requirement that consequential actions have human involvement.

This explains an asymmetry that would otherwise be arbitrary. A deterministic protective function — reactor trip, engineered safety feature actuation, protective relay operation, automatic load shedding — performs actions of the highest consequence without a Command Broker. Those actions are not below the line; asserting that they were would be false, and would break the criterion at §4.2. They are outside the obligation under the carve-out at §4.4, and the reason the carve-out is defensible is that a protective function's behavior is bounded by proof. What it will do is known before it does it. Judgment is not required where proof is available.

Where behavior cannot be bounded by proof, the corpus requires a qualified human to supply the judgment that proof cannot. That is the whole of the Command Broker's purpose, and it is the criterion against which any proposed alternative discharge of the obligation is to be evaluated.

Open at this version. Whether the authorization required by this section may, for a narrow and specified class of actions, be discharged in advance rather than in the moment is the subject of a determination in progress and is not settled here. §6.1 records the question and the constraints it operates under.

5. Worked Examples

These examples are illustrative and interchangeable. Sector-specific variants may be substituted for a given audience without altering the definition.

A supervised learning model performing vibration-based bearing failure prediction on a feedwater pump, deterministic within a stated input bound and formally testable as deployed, is outside the scope of Bright Line analysis under §4.4. If the same model is replaced by a generative system producing the same recommendations, the system enters scope, and each of its actions is then placed on consequence: a maintenance work order recommendation is below the line, and a recommendation that adjusts pump speed during operation is placed on what an incorrect adjustment could cause.

A generative optimization system on a natural gas processing train nominates operating points within an envelope whose setback from the safety instrumented system trip points was established by human authority in advance. The nomination of an operating point inside the envelope is one action. An adjustment of the envelope itself toward a trip point is a different action with a materially different consequence, and the two are placed separately. The second is above the line under any reading. The first is the action addressed by the mode-of-discharge determination at §6.1 (FD-BL-D1).

A generative system synthesizes plant state into a situational awareness display. The display changes no process parameter, and under a naive reading it is below the line. But the consequence of an action is assessed as the action functions in the plant, including its role in authorizing other actions. An operator who authorizes an above-line action against a corrupted or drifted synthesis is not blind; he is confident and wrong, and the human authorization the framework relies on has been hollowed out without any protective layer registering a fault. The placement of indication-integrity objects is qualified at §6.2 (FD-BL-D2).

A generative system recommends a change to a protective relay setting. The relay is a carved-out component under §4.4 and its own protective actions are outside the obligation. The recommendation is not. It is an in-scope component acting upon a carved-out component, it is placed on its own consequence, and an incorrect protective setting is among the most consequential outcomes available. A vendor characterizes a product as providing turnkey autonomous process optimization. The characterization has no effect on placement. Each action the product can take is classified on its consequence, and the actions above the line require the architecture at §4.5 regardless of what the product is marketed as doing.

6. Determined Items

The three items reserved at this section in prior versions are now determined. Each is carried by a dedicated instrument; the resolutions bind this definition by reference and are not restated here. The constraints this instrument placed on each determination (that a resolution may vary mode, not placement/granularity/scope) were honored by the issued instruments.

6.1 Mode of discharge for the authorization obligation — resolved; see FD-BL-D1

Whether the §4.5 authorization may be discharged in advance for a narrow class of actions is determined by **FD-BL-D1 (Mode of Discharge: Pre-Authorization by Verified Gatekeeper)**. Advance discharge is permitted only under R1 (the whole safety case in a deterministic, formally verified gatekeeper the proposer cannot modify or bypass), R2 (completeness and soundness proven over state, trajectory, and rate — bounding selection, not only admissibility), and R3 (envelope changes categorically excluded from advance discharge). The consequence classifier and the verified gatekeeper are two objects in series, not one. The composite-classification and third-tier routes remain foreclosed. Mode varies; placement, granularity, and scope do not.

6.2 Indication integrity — resolved; see FD-BL-D2

Whether a below-line placement is available to generative output that functions as the basis for an authorization is determined by **FD-BL-D2 (Indication Integrity)**. The disposition is a qualifier, not a placement change: such output must be traceable to independently verifiable values (I2) and must not be the sole basis for an above-line authorization (I3), with the obligation attaching by function (I1). The rule binds both the human-broker and the verified-gatekeeper cases.

6.3 Terminology reservation — resolved; see FD-EV

The two-sense collision on “envelope” is resolved by **FD-EV**, a Foundational Definition companion to this instrument and FD-LD. “Envelope” (and “pre-authorized envelope”) is reserved for the authorization sense; the operational sense — a region whose violation is a runtime event to detect and report — is renamed “operating band.” Conformance is staged per instrument revision, following the “Bright Line” reservation model.

7. Conformance

This instrument governs the definition of the Bright Line. Restatements shall be conformed at each instrument’s next revision, and the tier rule at §2.1 applies whether or not an instrument is named below.

Instruments presently varying from this definition, and the nature of the variance:

Command Broker Framework, Cross-Sector — states placement as an architectural layer boundary; does not carry action-level granularity; extends the enforcement requirement beyond the scope at §4.4. **Layer Decomposition and Determinism Terminology Annex** — maps the layer boundary onto the placement criterion. Its carve-out text is correct and is the source for §4.4. Anticipated re-designation as FD-LD.

DBA-EN Energy Sector Framework — states placement as a technology-class boundary and propagates that statement series-wide by restatement rather than by reference.

DBA-ES Sector Framework — restates the technology-class formulation; narrows the carve-out to enumerated roles.

DBA-VC Vendor Companion — conforming as written; requires citation of this instrument as the governing definition.

ISA Parent Framework, IEEE V&V Cognitive AI Framework, CB-Annex-Nuclear — state the obligation in terms coupled to §6.1, now determined by FD-BL-D1; conformed in the Pass 2 amendment set. The detailed diagnosis, replacement text, and sequencing for each instrument are carried in the separate conformance schedule and are not reproduced here.

8. Author Decisions Outstanding

Confirmation of the specific standard and clause references named at §4.4 against current editions. These are identified by ecosystem in this draft and should not issue as specific citations without verification.

Whether the conformance schedule is issued alongside this instrument or held for release on request.

9. Foundational References

Roxey, T.E. *Generative AI Risk in Industrial Control Systems: A Study for Critical Infrastructure Operators, Regulators, and Suppliers*. Version 29. Eclectic Technologies, 2026.

Command Broker Framework, Cross-Sector. Roxey, T.E. Eclectic Technologies, 2026.

Layer Decomposition and Determinism: The Two-Layer Architecture, the Determinism Boundary, and the Machine Learning Carve-Out. Roxey, T.E. Eclectic Technologies, 2026.

DBA-VC Vendor Companion. Roxey, T.E. Eclectic Technologies, 2026.

DBA-EN Energy Sector Framework; DBA-ES Sector Framework; DBA-GAS-FC2 Natural Gas Processing. Roxey, T.E. Eclectic Technologies, 2026.

Revision History

0.3 — August 2026. §6 open items collapsed to resolved pointers, all three now determined: §6.1 mode of discharge → FD-BL-D1 (R1–R3; two-object model; foreclosures retained); §6.2 indication integrity → FD-BL-D2 (I1–I3; human and machine cases unified); §6.3 terminology reservation → FD-EV (envelope reserved for the authorization sense, operational sense renamed “operating band”). Worked-example cross-references and the §7 conformance line updated to cite the issued determinations. No change to the definition (§§1–5).

0.2 — August 2026. Designated FD-BL and placed in the Foundational Definitions family. §2 restructured to carry the tier rule (foundational definitions take precedence over sector frameworks generally), the family membership test, the propagate-by-reference requirement, and the declaration convention. New §4.1 states what kind of thing the Bright Line is, in compliance with that convention; subsequent subsections renumbered. §4.4 carve-out: role-list narrowing struck, property test retained, scope statement added recording that the carve-out is not a grant of ungoverned autonomy and naming where obligations arise, with the coverage gap for machine-learned autonomous control acknowledged rather than papered; new provision that the carve-out does not extend to in-scope components acting upon carved-out components. §4.5 adds the verification rationale — the Command Broker is compensation for unverifiability — and derives the protective-function asymmetry from it. §6.1 constraint retained and its foreclosures named explicitly, with reasons and with the override of the nuclear annex owner’s stated preference recorded. §7 reduced to an operative register; diagnosis and sequencing moved to a separate conformance schedule. Worked example added for actions taken upon carved-out components.

0.1 — August 2026. Initial draft. Four-question frame; canonical definition; carve-out distinguished from the line; full reconciliation register.